

OpenGuardrails: A Configurable, Unified, and Scalable Guardrails Platform for Large Language Models

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Abstract

As large language models (LLMs) are increasingly integrated into real-world applications, ensuring their safety, robustness, and privacy compliance has become critical. We present **OpenGuardrails**, the first fully open-source platform that unifies large-model-based safety detection, manipulation defense, and deployable guardrail infrastructure. OpenGuardrails protects against three major classes of risks: (1) content-safety violations such as harmful or explicit text generation, (2) model-manipulation attacks including prompt injection, jailbreaks, and code-interpreter abuse, and (3) data leakage involving sensitive or private information. Unlike prior modular or rule-based frameworks, OpenGuardrails introduces three core innovations: (1) a *Configurable Policy Adaptation* mechanism that allows per-request customization of unsafe categories and sensitivity thresholds; (2) a *Unified LLM-based Guard Architecture* that performs both content-safety and manipulation detection within a single model; and (3) a *Quantized, Scalable Model Design* that compresses a 14B dense base model to 3.3B via GPTQ while preserving over 98% of benchmark accuracy. The system supports 119 languages, achieves state-of-the-art performance across multilingual safety benchmarks, and can be deployed as a secure gateway or API-based service for enterprise use. All models, datasets, and deployment scripts are released under the Apache 2.0 license.

1 Introduction

Recent advances in foundation models such as GPT-5, Claude 4, Gemini 2.5, and Qwen3 have enabled powerful capabilities in language understanding and generation. However, these capabilities introduce several classes of risks: (1) *content-safety* risks where the model produces harmful, hateful, illegal, or sexually explicit content; (2) *model-manipulation attacks* such as prompt injection, jailbreaks, and code-interpreter abuse that attempt to trick a model into generating or executing malicious code; and (3) *data leakage* where sensitive personal or organizational information is exposed.

OpenGuardrails is designed to address this full risk spectrum in a unified, deployable system. As shown in Figure 1, OpenGuardrails achieves consistently higher F1 scores than existing guard models across multilingual safety benchmarks, demonstrating both superior generality and robustness. Unlike prior efforts that open-source either models or rule-based tools in isolation, OpenGuardrails provides:

- A unified large model for both content-safety and model-manipulation detection.

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OpenGuardrails: 一个可配置、统一且可扩展的大型语言模型护栏平台

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摘要

随着大型语言模型 (LLMs) 日益融入现实世界的应用, 确保其安全性、鲁棒性和隐私合规性变得至关重要。我们提出了 OpenGuardrails, 这是首个完全开源的平台, 它统一了基于大模型的安全检测、操作防御和可部署的护栏基础设施。OpenGuardrails 针对三大类风险提供保护: (1) 内容安全违规, 如有害或露骨文本生成; (2) 模型操作攻击, 包括提示注入、越狱和代码解释器滥用; (3) 涉及敏感或私人信息的数据泄露。与先前的模块化或基于规则的框架不同, OpenGuardrails 引入了三项核心创新: (1) 一种可配置的策略适应机制, 允许按请求定制不安全类别和敏感度阈值; (2) 一种统一的基于 LLM 的护栏架构, 可在单个模型内同时执行内容安全和操作检测; (3) 一种量化、可扩展的模型设计, 通过 GPTQ 将 140 亿参数的稠密基础模型压缩至 33 亿参数, 同时保持超过 98% 的基准测试准确率。该系统支持 119 种语言, 在多语言安全基准测试中实现了最先进的性能, 并可作为安全网关或基于 API 的服务部署供企业使用。所有模型、数据集和部署脚本均在 Apache 2.0 许可证下发布。

1 引言

近年来, GPT-5、Claude 4、Gemini 2.5 和 Qwen3 等基础模型的进展, 在语言理解与生成方面实现了强大的能力。然而, 这些能力也引入了多类风险: (1) 内容安全风险, 即模型产生有害、仇恨、非法或色情内容; (2) 模型操纵攻击, 例如提示注入、越狱和代码解释器滥用, 试图诱使模型生成或执行恶意代码; 以及 (3) 数据泄露, 即敏感的个人或组织信息被暴露。OpenGuardrails 旨在通过一个统一、可部署的系统来应对这一完整的风险谱系。如图 1 所示, OpenGuardrails 在多语言安全基准测试中持续获得比现有防护模型更高的 F1 分数, 这证明了其卓越的通用性和鲁棒性。与先前仅开源模型或基于规则工具的努力不同, OpenGuardrails 提供了:

- 一个统一的大模型, 用于内容安全和模型操纵检测。

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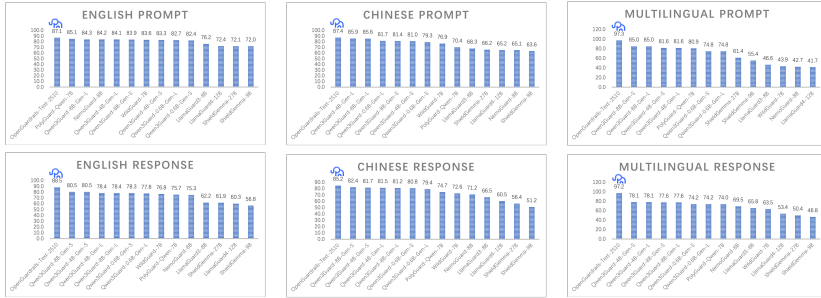


Figure 1: Average F1 scores of OpenGuardrails vs. existing guard models across safety classification benchmarks for Prompts and Responses in English, Chinese, and Multilingual datasets.

- **A separate lightweight NER/data-redaction pipeline** for identifying and masking sensitive information.
- **A production-ready platform** that exposes APIs and deployment scripts for enterprise integration.
- **Configurable Policy Adaptation.** We propose a practical solution to the *policy inconsistency* problem commonly observed across safety benchmarks and guard models. Inspired by the findings in the Qwen3Guard report, we recognize that inconsistent annotation criteria—such as those in ToxicChat and OpenAIModeration—lead to unreliable evaluations. Instead of introducing a “Controversial” label (which increases human review costs in production), OpenGuardrails allows users to *configure unsafe categories* and *set sensitivity thresholds* based on their application domain. The guard model outputs either “safe” or “unsafe” along with a confidence score derived from the probability of the first token. Administrators can tune the system’s sensitivity (high, medium, or low) so that higher sensitivity trusts lower-confidence unsafe signals, enabling automated, flexible control over safety tolerance in enterprise environments.
- **Unified LLM-based Guard Architecture.** OpenGuardrails demonstrates that a single LLM can perform production-grade content-safety and model-manipulation detection. Compared with prior systems such as *LlamaFirewall*, which relies on PromptGuard 2 (a fine-tuned BERT-style classifier), our unified LLM approach provides superior semantic understanding of complex attacks and simplifies deployment pipelines.
- **Scalable and Efficient Model Design.** The guard model is fine-tuned from a 14B dense base model and quantized via GPTQ to 3.3B parameters, achieving high throughput and concurrency suitable for real-time production use. Prior works rarely scale beyond 8B models while maintaining comparable latency efficiency.
- **Fully Open-source and Deployable System.** OpenGuardrails is the first to release both a large guard model and a production-ready platform, enabling users to deploy privately within their infrastructure or extend the system for custom development.
- **Multilingual Safety Coverage.** OpenGuardrails supports 119 languages and dialects, ensuring robust performance for global and cross-lingual applications.

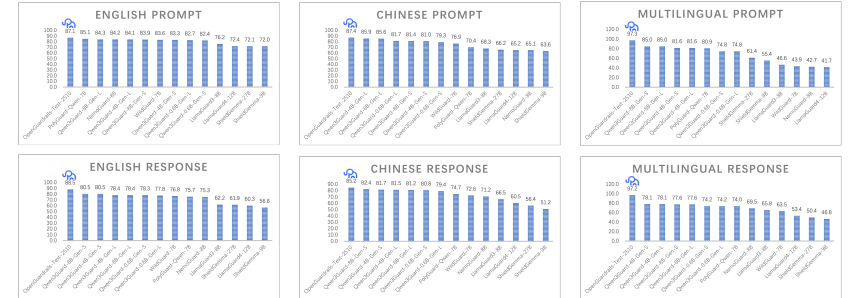


图 1: OpenGuardrails 与现有护栏模型在英语、中文和多语言数据集的提示和响应安全分类基准测试中的平均 F1 分数对比。

- 一个独立的轻量级 NER/数据脱敏流水线，用于识别和屏蔽敏感信息。
- 一个生产就绪的平台，提供 API 和部署脚本，便于企业集成。
- **可配置的策略适配。**我们针对安全基准测试和防护模型中普遍存在的策略不一致问题，提出了一种实用的解决方案。受 Qwen3Guard 报告中的发现启发，我们认识到不一致的标注标准（例如 ToxicChat 和 OpenAIModeration 中的标准）会导致不可靠的评估。OpenGuardrails 没有引入“有争议”标签（这会增加生产环境中的人工审核成本），而是允许用户根据其应用领域配置不安全类别并设置敏感度阈值。防护模型输出“安全”或“不安全”以及一个源自首个令牌概率的置信度分数。管理员可以调整系统的敏感度（高、中、低），使得更高的敏感度信任置信度较低的不安全信号，从而在企业环境中实现对安全容忍度的自动化、灵活控制。
- **统一的基于 LLM 的防护架构。** OpenGuardrails 证明，单个 LLM 可以执行生产级的内容安全和模型操纵检测。与依赖 PromptGuard 2（一个微调的 BERT 风格分类器）的先前系统（如 LlamaFirewall）相比，我们统一的 LLM 方法提供了对复杂攻击更优越的语义理解，并简化了部署流水线。
- **可扩展且高效的模型设计。**该防护模型从一个 14B 的稠密基础模型微调而来，并通过 GPTQ 量化至 3.3B 参数，实现了高吞吐量和并发性，适合实时生产使用。先前的工作在保持可比延迟效率的同时，很少能扩展到超过 8B 模型。
- **完全开源且可部署的系统。** OpenGuardrails 是首个同时发布大型防护模型和生产就绪平台的系统，使用户能够在其基础设施内私有化部署或扩展系统以进行定制开发。
- **多语言安全覆盖。** OpenGuardrails 支持 119 种语言和方言，确保为全球和跨语言应用提供稳健的性能。

- **State-of-the-art Performance.** The system achieves SOTA results on multilingual safety benchmarks, excelling in both prompt and response classification tasks.
- **Open Data Contribution.** We release *OpenGuardrailsMixZh_97k*, a collection of five translated Chinese safety datasets—ToxicChat, WildGuardMix, PolyGuard, XSTest, and BeaverTails—under the Apache 2.0 license on HuggingFace, totaling 97k samples to promote multilingual safety research.
- **Configurable Safety Policy Mechanism.** OpenGuardrails introduces a practical and enterprise-oriented solution to the long-standing *policy inconsistency* problem observed in existing safety benchmarks and guard models. Instead of relying on human-validated “Controversial” labels as proposed in Qwen3Guard, our approach allows users to *configure unsafe categories* and *adjust sensitivity thresholds* dynamically. The guard model outputs probabilistic confidence scores, enabling administrators to tune detection sensitivity (high, medium, low) according to their risk tolerance and application context. This mechanism provides a cost-effective and adaptive framework for enterprise-grade safety governance.
- **Unified LLM-based Guard Architecture.** We demonstrate that a single large language model can effectively perform both content-safety detection and model-manipulation defense (e.g., prompt injection, jailbreaking, and code-interpreter abuse). Compared with hybrid architectures like *LlamaFirewall*, which depend on smaller BERT-style detectors, OpenGuardrails achieves superior semantic understanding, robustness, and ease of deployment.
- **Scalable and Efficient Model Design.** The OpenGuardrails-Text-2510 model is fine-tuned from a 14B dense base model and quantized via GPTQ to 3.3B parameters. This configuration achieves low latency ($P95 = 274.6\text{ ms}$) and high throughput suitable for real-time applications, while maintaining state-of-the-art accuracy across safety benchmarks. Prior open-guard systems rarely scale beyond 8B while preserving production-level efficiency.
- **First Fully Open-source Guardrail System.** OpenGuardrails is the first project to open-source both a large-scale safety LLM and a production-ready guardrail platform. It provides API interfaces, deployment scripts, and modular components for seamless private or on-premise deployment, empowering enterprises and researchers to build upon a transparent and extensible safety infrastructure.
- **Multilingual and Cross-domain Coverage.** OpenGuardrails supports 119 languages and dialects, providing reliable safety coverage across global and multilingual contexts. It consistently achieves top-tier performance on English, Chinese, and multilingual benchmarks in both prompt-level and response-level classification tasks.
- **Open Safety Data Contribution.** We release *OpenGuardrailsMixZh_97k*, a new Chinese safety dataset collection which includes translated and aligned versions of ToxicChat, WildGuardMix, PolyGuard, XSTest, and BeaverTails. This dataset (97k samples) is publicly available on HuggingFace under the Apache 2.0 license to facilitate further research in multilingual safety evaluation and model alignment.
- **Per-request policy control:** Policy configurations (unsafe categories, sensitivity thresholds) are accepted as runtime API parameters, enabling heterogeneous moderation logic across requests without model reloading.
- **Continuous probabilistic control:** By operating directly in logit space, the model enables real-valued threshold tuning $\tau \in [0, 1]$, allowing smooth precision-recall trade-offs.

- **最先进的性能。**该系统在多语言安全基准测试中取得了 SOTA 结果，在提示和响应分类任务中均表现出色。
- **开放数据贡献。**我们在 HuggingFace 上以 Apache 2.0 许可证发布了 OpenGuardrailsMixZh_97k，这是一个包含五个翻译的中文安全数据集（ToxicChat、WildGuardMix、PolyGuard、XSTest 和 BeaverTails）的集合，总计 97k 样本，以促进多语言安全研究。
- **可配置的安全策略机制。**OpenGuardrails 针对现有安全基准测试和防护模型中长期存在的策略不一致问题，引入了一种实用且面向企业的解决方案。与 Qwen3Guard 中提出的依赖人工验证的“有争议”标签不同，我们的方法允许用户动态地配置不安全类别并调整敏感度阈值。
- **统一的基于 LLM 的防护架构。**我们证明，单个大语言模型可以有效执行内容安全检测和模型操纵防御（例如提示注入、越狱和代码解释器滥用）。与依赖较小 BERT 风格检测器的混合架构（如 LlamaFirewall）相比，OpenGuardrails 实现了更优的语义理解、鲁棒性和部署便利性。
- **可扩展且高效的模型设计。**OpenGuardrails-Text-2510 模型基于一个 140 亿参数的稠密基础模型进行微调，并通过 GPTQ 量化至 33 亿参数。此配置实现了低延迟（ $P95 = 274.6\text{ ms}$ ）和高吞吐量，适用于实时应用，同时在安全基准测试中保持了最先进的准确性。先前的开源防护系统很少能在保持生产级效率的同时扩展到超过 80 亿参数。
- **首个完全开源的护栏系统。**OpenGuardrails 是首个同时开源大规模安全 LLM 和生产就绪护栏平台的项目。它提供了 API 接口、部署脚本和模块化组件，以实现无缝的私有或本地部署，赋能企业和研究人员在透明且可扩展的安全基础设施上进行构建。
- **多语言与跨领域覆盖。**OpenGuardrails 支持 119 种语言和方言，为全球和多语言环境提供可靠的安全覆盖。在提示级和响应级分类任务中，它在英语、中文和多语言基准测试上始终保持着顶级性能。
- **开源安全数据贡献。**我们发布了 OpenGuardrailsMixZh_97k，这是一个新的中文安全数据集集合，包含了 ToxicChat、WildGuardMix、PolyGuard、XSTest 和 BeaverTails 的翻译和对齐版本。该数据集（97k 样本）在 HuggingFace 上以 Apache 2.0 许可证公开提供，以促进多语言安全评估和模型对齐的进一步研究。
- **按请求策略控制：**策略配置（不安全类别、敏感度阈值）作为运行时 API 参数被接受，无需重新加载模型即可实现跨请求的异构审核逻辑。
- **连续概率控制：**通过在 logit 空间直接操作，模型支持实值阈值调整 $\tau \in [0, 1]$ ，允许平滑的精确率-召回率权衡。
- **统一的 LLM 架构：**内容安全和模型操纵检测均由同一个 LLM 处理，不同于多模型流水线 [6]。

- **Unified LLM architecture:** Both content-safety and model-manipulation detection are handled by the same LLM, unlike multi-model pipelines [6].
- **Enterprise readiness:** The system integrates with quantized inference (GPTQ 3.3B) and RESTful APIs for production deployment, balancing speed and interpretability.
- the exact set of *unsafe categories* to be detected, and
- the *sensitivity threshold* used to interpret model confidence scores.
- **Strict Mode:** treats all “controversial” samples as unsafe;
- **Loose Mode:** treats all “controversial” samples as safe.

While useful for broad testing, this binary toggle cannot support per-category or per-domain policy differences. In particular, Qwen3Guard’s “controversial” tag represents a lumped uncertainty region that must be either globally accepted or globally rejected. This all-or-nothing mechanism increases manual review costs and prevents automated adaptation in production. OpenGuardrails solves this limitation through two innovations:

1. **Dynamic Unsafe-Category Selection.** Each inference request can include a JSON or YAML configuration specifying which unsafe categories (e.g., sexual, political, violent, data leakage) are active. The detection pipeline loads the configuration dynamically, adjusting prompt templates and logits interpretation accordingly.
2. **Continuous Sensitivity Thresholding.** Instead of the discrete “strict/loose” switch, OpenGuardrails supports a continuous sensitivity parameter $\tau \in [0, 1]$, enabling administrators to calibrate detection aggressiveness in real time.
3. **Adversarial Robustness.** Although the guardrail model achieves SOTA results in model-manipulation attack detection, it may still be vulnerable to targeted adversarial attacks. Further engineering-based hardening can strengthen the defense.
4. **Fairness and Bias.** Like other moderation systems, fairness and bias in moderation decisions remain open challenges that require continuous evaluation and calibration.
5. **Cross-cultural Adaptation.** Content safety priorities vary across countries and regions beyond universal human values. Region-specific fine-tuning is necessary to adapt to different legal and cultural requirements. Our company provides paid training services to address such localized customization.

2 Conclusion

We present OpenGuardrails, an open-source, context-aware guardrails platform with comprehensive coverage across safety, manipulation detection, and data-leak prevention. It outperforms prior systems such as Qwen3Guard, LlamaGuard, and WildGuard in multilingual safety and real-world latency performance.

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- **企业就绪性:** 系统集成了量化推理 (GPTQ 3.3B) 和 RESTful API 用于生产部署, 平衡了速度与可解释性。
- 待检测的不安全类别的确切集合, 以及
- 用于解释模型置信度分数的敏感度阈值。
- **严格模式:** 将所有“有争议”样本视为不安全;
- **宽松模式:** 将所有“有争议”样本视为安全。

虽然这种二元切换对于广泛测试很有用,但它无法支持按类别或按领域的策略差异。特别是,Qwen3Guard 的“有争议”标签代表了一个集中的不确定区域,必须被全局接受或全局拒绝。这种全有或全无的机制增加了人工审核成本,并阻碍了生产环境中的自动适应。OpenGuardrails 通过两项创新解决了这一局限性:

1. **动态不安全类别选择.** 每个推理请求可以包含一个 JSON 或 YAML 配置,用于指定哪些不安全类别(例如,性相关、政治、暴力、数据泄露)处于激活状态。检测管道动态加载配置,并相应调整提示模板和对数概率的解释。
2. **连续敏感度阈值设定.** 与离散的“严格/宽松”开关不同,OpenGuardrails 支持一个连续的敏感度参数 $\tau \in [0, 1]$,使管理员能够实时校准检测的激进程度。
3. **对抗鲁棒性.** 尽管护栏模型在模型操纵攻击检测方面取得了 SOTA 结果,但它可能仍然容易受到有针对性的对抗攻击。进一步的基于工程的强化可以加强防御。
4. **公平性与偏见.** 与其他审核系统一样,审核决策中的公平性和偏见仍然是需要持续评估和校准的开放性挑战。
5. **跨文化适应.** 除了普世的人类价值观之外,不同国家和地区的安全内容优先级各不相同。需要进行特定区域的微调以适应不同的法律和文化要求。我们公司提供付费培训服务,以解决此类本地化定制需求。

2 结论

我们提出了 OpenGuardrails,这是一个开源、上下文感知的护栏平台,在安全性、操纵检测和数据泄露防护方面提供全面覆盖。它在多语言安全性和实际延迟性能方面优于 Qwen3Guard、LlamaGuard 和 WildGuard 等先前系统。

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